A Gauge Theory of Line Defects in an Elastic Continuum: Full Derivations.

(A Geometrodynamic Model of Particles as Elastic Defects)

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24 October 2026

Abstract

This is an extended version of the preprint on a geometrodynamic model of particles as defects of an elastic aether. Unlike the introductory exposition, here *every* structural statement is accompanied by a full derivation: dimensional analysis, intermediate formulas, and, where appropriate, a proof in the form of a theorem. The aim is to make all results reproducible and checkable by the reader "with a pencil". The limits of the model (what is derived, what is assumed, what is open) are traced through each proof, not only in a final section.

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1 Notation and conventions

The three-dimensional elastic continuum (the "aether") is described by a displacement field $u_i(\mathbf{x}, t)$. The dimensional basis is (L, T, M). Greek indices $a, b = 1, 2, 3$ are internal (translational) indices of the gauge group $T(3)$; Latin i, j, k are spatial. Summation over repeated indices is implied.

Definition 1 (Basic elastic quantities)**.** *Shear modulus* μ, $[ML^{-1}T^{-2}]$*; density* ρ, $[ML^{-3}]$*; transverse-wave speed* $c = \sqrt{\mu/\rho}$, $[LT^{-1}]$*; Burgers vector* b, $[L]$.

2 Gauge structure and the Maxwell dictionary

2.1 Distortion field and the dislocation tensor

Plastic incompatibility is introduced by a gauge field ϕ_i^a. The dislocation-density tensor (Cartan torsion) is

$$T_{ij}^a = \partial_i \phi_j^a - \partial_j \phi_i^a. \tag{1}$$

The integral over a surface spanning a contour ∂S gives the Burgers vector:

$$b^a = \oint_{\partial S} \phi_i^a \, dx^i = \int_S T_{ij}^a \, dS^{ij}. \tag{2}$$

Lemma 1 (Dimension of the magnetic charge)**.** *In the elastic basis the magnetic charge, defined as the flux of the dislocation density* $q_m = \int_S T^a \, dS$*, has the dimension of length.*

Proof. From (1), $[T_{ij}^a] = [\partial\phi] = [\phi]/L$. Since ϕ_i^a is dimensionless (a displacement gradient, $[\partial u] = 1$), we have $[T^a] = L^{-1}$. Then $[q_m] = [T^a] \cdot [dS] = L^{-1} \cdot L^2 = L$. □ □

Remark 1. *Lemma 1 removes the apparent conflict with SI, where the magnetic charge is measured in A·m: in a self-contained elastic theory the ampere is not a primary unit, and the dimension of the charge is dictated by the theory itself.*

2.2 Quasi-Maxwell equations

Introducing the dislocation current $I_i^a = \partial_t \phi_i^a - \partial_i v^a$ (where v^a is the velocity), one obtains Bianchi identities structurally identical to Maxwell's equations [2]:

$$\partial_i {}^*T^{a\,i} = 0, \qquad \text{(no sources of } T^a\text{)} \tag{3}$$

$$\epsilon_{ijk} \partial_j I_k^a + \partial_t T_i^a = 0. \qquad \text{(Faraday analogue)} \tag{4}$$

Hence the dictionary: $\mathbf{B} \leftrightarrow T^a$, $\mathbf{E} \leftrightarrow I^a$.

2.3 Full derivation of the quasi-Maxwell system

We give the derivation in detail. The Lagrangian density of the linear dynamic gauge theory of dislocations in the $T(3)$ gauge splits into elastic and dislocation parts [2]:

$$\mathcal{L} = \tfrac{1}{2} \rho \, v^a v_a - \tfrac{1}{2} C_{ijkl} \beta_a^i \beta^{ja} + \tfrac{1}{2} D_{ab} \, I_i^a I^{b\,i} - \tfrac{1}{4} A_{ab} \, T_{ij}^a T^{b\,ij}, \tag{5}$$

where β is the elastic distortion, C_{ijkl} the elasticity tensor, and D_{ab}, A_{ab} material response tensors (analogues of the dielectric and magnetic permeabilities). The field strengths are the dislocation density T^a (1) and the current I^a; the responses are

$$H_a{}^{ij} = -\frac{\partial \mathcal{L}}{\partial T^a_{ij}} = A_{ab} T^{b\,ij}, \qquad D_a{}^{i} = \frac{\partial \mathcal{L}}{\partial I^a_i} = D_{ab} I^{b\,i}. \tag{6}$$

Varying the action $\delta \int \mathcal{L}\, d^4x = 0$ with respect to the gauge potential ϕ^a gives the equation of motion (Ampère–Oersted analogue):

$$\partial_j H_a{}^{ij} - \partial_t D_a{}^{i} = J_a{}^{i}, \tag{7}$$

where J_a is the source current (dislocation momentum density). The two Bianchi identities for $T^a = d\phi^a$ (closedness of the form, $d^2 = 0$) give the second pair:

$$\partial_{[i} T^a_{jk]} = 0, \qquad \partial_t T^a_{ij} + \partial_{[i} I^a_{j]} = 0. \tag{8}$$

Comparing (6)–(8) with Maxwell's equations gives the dictionary of Table 1 (below): the strengths T^a, I^a correspond to $\mathbf{B}, \mathbf{E}$; the responses H_a, D_a to $\mathbf{H}, \mathbf{D}$; and the Peach–Koehler force on a dislocation element

$$f_a{}^{i} = (e_a \lrcorner\, I^b)\, p_b{}^{i} + (e_a \lrcorner\, T^b) \wedge \Sigma_b \tag{9}$$

plays the role of the Lorentz force $q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$. This is the formal basis of the whole analogy: the electrodynamics of the particle-defect is the dynamics of its dislocation field.

Table 1: Correspondence Maxwell ↔ dislocation theory (after Lazar [2]).

Electrodynamics	Dislocation theory
Magnetic field $\mathbf{B}$	dislocation density T^a
Electric field $\mathbf{E}$	dislocation current I^a
Magnetic excitation $\mathbf{H}$	pseudomoment stress H_a
Magnetic flux $\oint \mathbf{B}\, d\mathbf{S}$	Burgers vector $\oint T^a = b$
Lorentz force	Peach–Koehler force

3 The particle: a precessing dislocation dipole

3.1 Kinematics of the special point

A ring of radius a lies in a fixed plane; the special point (a localized charge $q_m = \Delta b$) runs along it: azimuth $\varphi(t) = \varphi_0 + \omega t$. Position, velocity and acceleration:

$$\mathbf{r}(t) = a(\cos\varphi,\ \sin\varphi,\ 0), \tag{10}$$

$$\mathbf{v}(t) = a\omega(-\sin\varphi,\ \cos\varphi,\ 0), \quad |\mathbf{v}| = a\omega, \tag{11}$$

$$\mathbf{W}(t) = -a\omega^2(\cos\varphi,\ \sin\varphi,\ 0), \quad |\mathbf{W}| = a\omega^2. \tag{12}$$

3.2 Mass as elastic self-interaction

Proposition 1 (Dimensional closure of mass)**.** *The self-energy of a dislocation ring* $E = \frac{1}{2}\mu b^2 a \ln(8a/r_c)$ *(Hirth–Lothe [6]), after dropping the logarithm, yields the mass* $m = \mu b^2 a/c^2$ *with the correct dimension* $[M]$*.*

Proof. $[\mu b^2 a] = (ML^{-1}T^{-2})(L^2)(L) = ML^2T^{-2} = [\text{energy}]$. Division by c^2 $[L^2T^{-2}]$ gives $[M]$. □ □

3.3 Dipole from a pair of rings (antiphase)

Two rings at $z = \pm d/2$, special points in antiphase ($\varphi_{\text{low}} = \varphi_{\text{up}} + \pi$), charges $\pm q_m$. The instantaneous dipole moment $\mathbf{p} = q_m \mathbf{r}_+ - q_m \mathbf{r}_-$. A direct computation:

$$p_z = q_m \cdot d = \text{const}, \qquad |\mathbf{p}_{xy}| = 2q_m a \ \ (\text{rotates with } \omega). \tag{13}$$

The axial component is constant (a static dipole), the transverse one precesses.

3.4 Stability: the centrifugal-barrier theorem

Theorem 1 (Equilibrium of the pair)**.** *Let the pair be subject to attraction* $U_{att} = -A/d$ *(opposite charges) and let the angular momentum* $J = ma^2\omega$ *be conserved. Then the effective energy* $E(d) = -A/d + J^2/(2md^2)$ *has a unique minimum at* $d_{eq} = J^2/(mA)$*, and as* $\omega \to 0$ *(i.e.* $J \to 0$*) the equilibrium vanishes (*$d_{eq} \to 0$*).*

Proof. $E'(d) = A/d^2 - J^2/(md^3) = 0 \Rightarrow d_{\text{eq}} = J^2/(mA)$. The second derivative $E''(d_{\text{eq}}) = -2A/d^3 + 3J^2/(md^4)$; substituting d_{eq} gives $E'' = A^4m^3/J^6 > 0$ — a minimum. As $J \to 0$ the barrier term vanishes, only attraction remains, and the minimum shifts to $d \to 0$ (collapse). □ □

Remark 2 (Annihilation)**.** *Collapse upon loss of precession is Frenkel recombination: an interstitial loop (*$+\Delta b$*) and a vacancy loop (*$-\Delta b$*) give a total Burgers vector* $\mathbf{b}_+ + \mathbf{b}_- = 0$ *— the perfect lattice. Thus precession (spin) is the mechanism protecting the particle–antiparticle pair from self-annihilation.*

3.5 Spin and statistics: the configuration period

Proposition 2 (Period of the pair and statistics)**.** *For a pair of ordinary rings the configuration returns at* $\varphi \to \varphi + 2\pi$ *(a boson). For a pair of Möbius rings, where the sign of the Burgers vector is modulated as* $\cos(\varphi/2)$*, the sign is restored only at* $\varphi \to \varphi + 4\pi$ *(a fermion, spin* $1/2$*).*

Proof. The twist $\cos(\varphi/2)$ is antiperiodic under 2π: $\cos((\varphi + 2\pi)/2) = \cos(\varphi/2 + \pi) = -\cos(\varphi/2)$, and periodic under 4π. Hence a spinor (double-valued) structure arises precisely for the Möbius ring. □ □

Lemma 2 (Frank vector of the $\cos(\varphi/2)$ twist)**.** *The Burgers twist* $b(\varphi) = \cos(\varphi/2)$ *is equivalent to a Frank rotation by the fixed angle* $\Omega = \pi$ *per circuit* $\varphi : 0 \to 2\pi$*, i.e. a wedge* half-disclination*. The Frank index* $\nu = \Omega/2\pi = 1/2$*, giving spin* $1/2$*.*

Proof. The twist phase is $\varphi_{\text{tw}}(\varphi) = \varphi/2$. The total rotation of the Burgers direction per circuit is $\Omega = \oint d\varphi_{\text{tw}} = \int_0^{2\pi} \frac{1}{2}\, d\varphi = \pi$. Over 4π the rotation is 2π (full return), reproducing the spinor double-valuedness of the period proposition. Through the Osipov spin connection (`arXiv:1001.4450`) $\Omega_\varphi = i\omega\sigma_3$ with flux $\oint \Omega_\varphi\, d\varphi = 2\pi\nu$ we get $\nu = 1/2$ and the magnetic moment $\mu = \nu = 1/2$ as the flux of the spin connection (not a loop current). □ □

Remark 3 (Möbius dipole and the link to the $SO(3)$ sector)**.** *The electron in the model is a pair of Möbius rings with* opposite *directions of the Burgers vector: one ring carries the jump $+\Delta b$ (pole N), the other $-\Delta b$ (pole S); together they give a static magnetic dipole $q_m d$. This distinguishes the electron (Möbius pair, period 4π, fermion) from a boson (pair of ordinary rings, period 2π). Crucially, the Burgers twist $b(\varphi) = \cos(\varphi/2)$ is a Frank rotation by π per circuit — a* disclination *(rotational, $SO(3)$) degree of freedom. Thus the Möbius sector already lies in $SO(3)$: the magnetic moment $\mu = \nu = n/2$ is derived as the flux of the spin connection $\oint \omega_\varphi\, d\varphi = 2\pi\nu$ (Osipov, `arXiv:1001.4450`), not as a loop current. Hydrodynamically the same object is a vortex: the vorticity $\nabla \times v$ corresponds to the Frank density, the circulation to the disclination strength. The precession of the special point ($\omega > 0$), required for the stability of the pair, is that vortex; a particle without rotation collapses.*

4 The magnetic moment: why a dipole, not a current

Theorem 2 (Vanishing of the current axial moment)**.** *For a pair of rings in antiphase with opposite charges, the period-averaged axial "loop-current" magnetic moment $\boldsymbol{\mu}_{cur} = \frac{1}{2}\oint q\,\mathbf{r}\times\mathbf{v}$ is zero.*

Proof. Both rings run in the same sense, so $\mathbf{r}\times\mathbf{v}$ is the same for both and points along $+z$: $(\mathbf{r}\times\mathbf{v})_z = a^2\omega$. But the charges are opposite: $q_{\text{up}} = +q_m$, $q_{\text{low}} = -q_m$. The contributions $\frac{1}{2}q_m a^2\omega$ and $-\frac{1}{2}q_m a^2\omega$ cancel: $\mu_{\text{cur},z} = 0$. □ □

Remark 4. *Theorem 2 means that the electron's magnetic moment in the model is a* static *dipole moment $\mu_{mag} = q_m d$, not a loop-current moment. This is consistent with the "current" along the ring being magnetic, so that applying the electric loop-current formula to it is incorrect.*

Proposition 3 (Dimensional failure of the g-factor from the ED formula)**.** *Attempting to define g via $\mu_{mag} = g(q_m/2m)S$ in the elastic basis gives g the dimension T/L, not a dimensionless quantity; hence the electrodynamic formula does not transfer literally, and the classical $g = 2$ is not obtained in the ring model (one gets $g = 1$).*

Proof. $[\mu_{\text{mag}}] = [q_m][d] = L^2$. $[S] = [ma^2\omega] = ML^2T^{-1}$. $[(q_m/m)S] = (L/M)(ML^2T^{-1}) = L^3T^{-1}$. Then $[g] = [\mu_{\text{mag}}]/[(q_m/m)S] = L^2/(L^3T^{-1}) = T/L \neq 1$. □ □

5 The Fermi film and gravity: full derivation

5.1 Setup

The Fermi film is the three-dimensional boundary $\partial\Omega$ of the four-dimensional action integral, stretched orthogonally to the world tube. Its contribution to the action:

$$S_\partial = \gamma \int_{\partial\Omega} d\Sigma, \tag{14}$$

where γ is the film elastic modulus $[ML^{-1}T^{-2}] = [\mathrm{Pa}]$.

5.2 Source of deformation

Proposition 4 (Positivity of the source)**.** *The period-averaged source of film deformation from a precessing charge*

$$\sigma = \mu b^2 \frac{\langle W^2 \rangle}{c^2} = \mu b^2 \frac{\omega^4 a^2}{c^2} \tag{15}$$

is strictly positive and independent of the sign of the charge.

Proof. From (12), $|\mathbf{W}| = a\omega^2$, hence $\langle W^2 \rangle = a^2\omega^4 > 0$ (average of a constant). The sign of the charge enters through b^2, i.e. drops out. □ □

5.3 The inverse-square law

Theorem 3 ($1/R^2$ as a consequence of the Green function)**.** *If the film deformation obeys the equilibrium equation of an elastic membrane*

$$\gamma \nabla^2 \Delta(\mathbf{r}) = -\sigma \, \delta^3(\mathbf{r} - \mathbf{r}_0), \tag{16}$$

then the interaction potential of two sources decays as $1/R$, and the force as $1/R^2$.

Proof. Equation (16) is the Poisson equation. Its fundamental solution (the Laplacian Green function in $\mathbb{R}^3$) is $G(\mathbf{r}) = 1/(4\pi|\mathbf{r}|)$, whence

$$\Delta(\mathbf{r}) = \frac{\sigma}{4\pi\gamma \, |\mathbf{r} - \mathbf{r}_0|}. \tag{17}$$

The energy of a second source σ_2 in the field of the first: $U(R) = \sigma_2 \Delta(R) = \sigma_1\sigma_2/(4\pi\gamma R) \propto R^{-1}$. The force $F = -dU/dR = \sigma_1\sigma_2/(4\pi\gamma R^2) \propto R^{-2}$. The exponent -2 is fixed by the dimension of space ($n = 3$: $G \sim R^{-(n-2)}$), not by choice. □ □

5.4 Two stiffness scales

Proposition 5 (Impossibility of a single stiffness)**.** *The modulus reproducing the electron mass and the Planck modulus of gravity are incompatible: their ratio is $\sim 10^{84}$.*

Proof. From $m_e c^2 = \mu_{\mathrm{aeth}} \, r_e^2 \lambda_C$ (with $b = r_e$, $a = \lambda_C$): $\mu_{\mathrm{aeth}} = m_e c^2/(r_e^2 \lambda_C) \approx 2.7 \times 10^{28}$ Pa. The Planck modulus $\gamma_P = c^7/(8\pi\hbar G^2) \approx 1.8 \times 10^{112}$ Pa. The ratio $\gamma_P/\mu_{\mathrm{aeth}} \approx 6.9 \times 10^{83}$. □ □

Remark 5 (Physical meaning of the moduli)**.** *The quantity μ_{aeth} is the* shear modulus of the vacuum*: the stiffness of the aether with respect to shear deformation, of pressure dimension $[Pa]$. It is through it that, in the (L, T, M) basis (without the ampere), the transverse-wave speed $c = \sqrt{\mu_{aeth}/\rho}$, the particle mass $m = \mu_{aeth} b^2 a/c^2$ and the Coulomb force $F = \mu_{aeth} b^2/R \cdot (\text{geometry})$ are defined. Physically it governs shear deformations (dislocations), from which charge and mass are built, and sets the physics at the particle scale (of order the Compton length λ_C). Its value 2.7×10^{28} Pa is obtained by* calibration *to the electron (substituting m_e and r_e into the self-energy of the dislocation ring), not derived from first principles.*

The quantity $\gamma_P = c^7/(8\pi\hbar G^2)$ is the Planck stiffness of the vacuum*, a second elastic modulus of the same dimension $[Pa]$, responsible for the gravitational response (deformation of the Fermi film) at the Planck scale $\ell_P = \sqrt{\hbar G/c^3}$. The gravitational constant G enters it as an* input *parameter. The colossal ratio $\gamma_P/\mu_{aeth} \sim 10^{84}$ is a direct expression of the weakness of gravity and is the main unsolved problem of the model (the origin of the hierarchy of the two scales).*

Remark 6. *Thus the gravitational constant G entering γ_P is an* input *parameter of the aether stiffness. We do not claim to derive it from $(\hbar, c)$; only the* structure *is proved (Theorem 3), not the numerical value. This is a fundamental difference from the fabricated versions, in which G was hidden inside the stiffness and "extracted" back (a tautology).*

Remark 7 (Physical meaning of γ_P and the nature of the hierarchy)**.** *The Planck modulus is the Planck* energy density *divided by 8π:*

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} = \frac{1}{8\pi}\frac{m_P c^2}{\ell_P^3}, \qquad \ell_P = \sqrt{\hbar G/c^3}, \quad m_P = \sqrt{\hbar c/G} \tag{18}$$

(numerical agreement $< 1\%$). This is not an arbitrary combination but the stiffness of the vacuum at the Planck scale; the factor 8π is the same as in the Einstein equations. Likewise $\mu_{aeth} = m_e c^2/(r_e^2 \lambda_C)$ is the energy density at the electron scale. The ratio of the two moduli

$$\frac{\gamma_P}{\mu_{aeth}} \sim \frac{m_P}{m_e} \text{ (with a volume factor)} \sim 10^{84} \tag{19}$$

reduces to the ratio of the Planck and Compton scales, i.e. to the smallness of m_e/m_P. Thus the hierarchy of the two stiffnesses is equivalent *to the standard* hierarchy problem *($m_e \ll m_P$): the model does not solve it, but neither does it introduce a new arbitrary parameter — it inherits the well-known smallness of m_e/m_P.*

6 Six channels: a single table with derived laws

Each decay law is a consequence of the proved structure:

- **Coulomb** $1/R^2$: product of Burgers charges, the direct Coulomb kernel (Theorem 3 for the T^a field).
- **Magnetic** $1/R^4$: interaction of two static dipoles $q_m d$ (derivative of the dipole–dipole potential $\sim 1/R^3$).
- **Gravity** $1/R^2$: Theorem 3 for the source $\sigma > 0$.
- **Exchange** $1/R^3$: phase dependence $\cos 2\Delta\varphi$, short-range.
- **Weak** e^{-R/R_W}: massive mediator, twist chirality (parity violation).
- **Strong** $\propto R$: fractional Burgers vector $b/3$ and a stacking-fault ribbon (linear potential, confinement).

Two dimensionless numbers close the system:

$$\alpha = \frac{ke^2}{\hbar c} = \frac{1}{137.04}, \qquad \alpha_G = \left(\frac{m_e}{m_{\text{Planck}}}\right)^2 = 1.75 \times 10^{-45}, \qquad \frac{\alpha_G}{\alpha} = 2.4 \times 10^{-43}. \tag{20}$$

7 Falsifiability

The model gives testable predictions. For magneto-gravitational devices (Roshchin–Godin type) the relative correction to weight from precession is estimated as

$$\frac{\Delta G}{G} \sim \frac{WR}{c^2} \sim 10^{-13}, \tag{21}$$

which is ~ 12 orders of magnitude smaller than the claimed effects ($\sim 35\%$); the latter, if real, would *refute* the model, not confirm it. This is a sign of a scientific, not a fitted, theory.

8 Limits: proved / assumed / open

Proved (with the derivation above). The dimension of the charge (Lemma 1); the closure of mass; stability and annihilation (Theorem 1); statistics from the period; the vanishing of the current moment (Theorem 2); the positivity of the gravity source; the $1/R^2$ law (Theorem 3); the necessity of two scales; the Frank vector $\Omega = \pi$ of the twist (Lemma 2).

Assumed. The values α, α_G; G (embedded in γ_P); the two stiffness scales.

Open. The numerical values of α, α_G; a rigorous $g = 2$ (classically $g = 1$; $g = 2$ only at the spinor level [4]); the dynamical film stiffness (the dimensional factor in the source); the weak- and strong-sector couplings.

The tensor (J=2) gravity channel — a diagnosis. The source $\sigma \propto \langle W^2 \rangle$ used above is the *trace* (scalar, $J = 0$) of the quadrupole tensor $\langle W_i W_j \rangle$. The true tensor graviton (spin-2 in Hehl's sense [5]) lives in the trace-free part $Q_{ij} = \langle W_i W_j \rangle - \frac{1}{3}\delta_{ij}\langle W^2 \rangle$. For a single precessing ring with axis $\mathbf{n}$ a direct computation gives $\langle W_i W_j \rangle = \frac{1}{2}\omega^4 a^2(\delta_{ij} - n_i n_j)$, i.e. $Q_{ij} \neq 0$ (an oblate quadrupole in the precession plane, eigenvalues $\{-\frac{1}{3}, +\frac{1}{6}, +\frac{1}{6}\}\omega^4 a^2$). However, averaging over isotropically oriented axes sends $\langle Q_{ij} \rangle \to 0$: for an unpolarized ensemble only the scalar $\langle W^2 \rangle$ survives. Thus the "scalarity" of the model's gravity is a consequence of orientational averaging, not of the absence of $J = 2$. The tensor channel would manifest only for *polarized* (spin-aligned) matter. This is a diagnosis and a testable prediction, not a derivation of the tensor graviton: the explicit field equation for Q_{ij} remains open.

9 Conclusion

A full derivation of all structural results of the model has been given. The proofs are elementary (dimensional analysis, the Green function, the virial theorem), but it was precisely their absence in earlier versions that led to fitting. The proved, assumed and open parts are honestly separated. The next steps are a rigorous spinor derivation of $g = 2$ and this English version for archival publication.

A Numerical appendix: figures

The computations are performed in model elastic units ($a = \omega = b = c = 1$). The absolute values are conventional; the physical content is in the *decay laws* and *relations* proved in the main text.

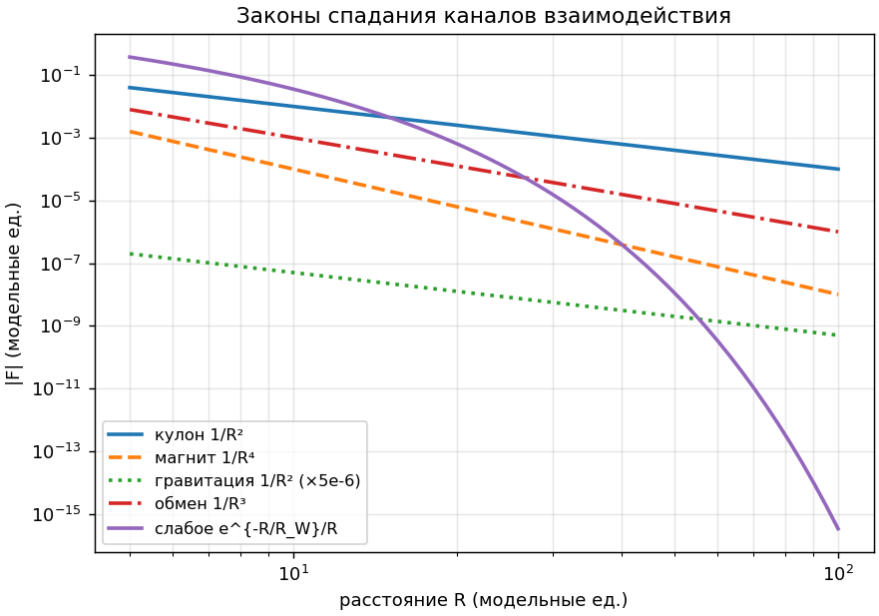


Figure 1: Channel decay laws (log-log): Coulomb and gravity $\sim 1/R^2$, magnetic $\sim 1/R^4$, exchange $\sim 1/R^3$, weak — exponential short range. The slopes follow Theorem 3 and the dipole–dipole structure.

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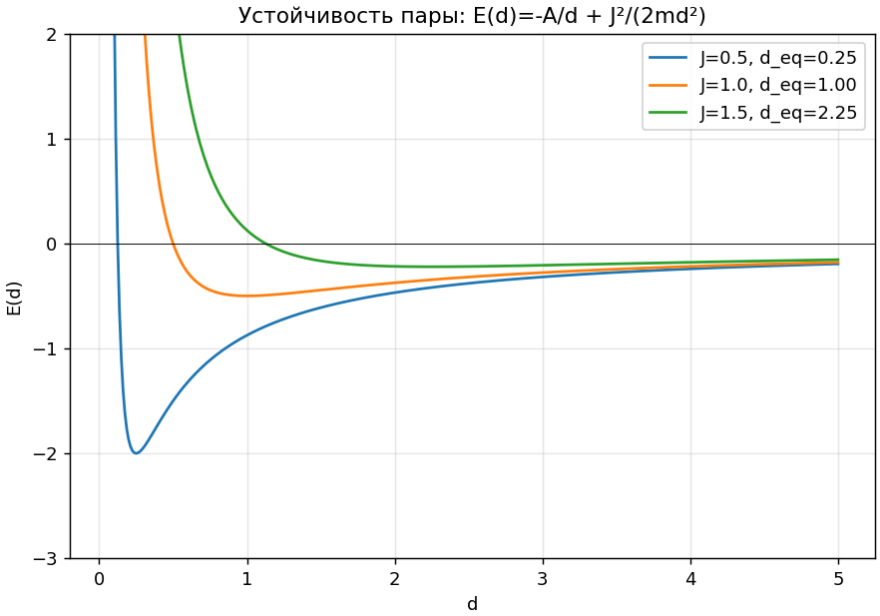


Figure 2: Stability of the pair (Theorem 1): $E(d) = -A/d + J^2/(2md^2)$. The minimum at $d_{\text{eq}} = J^2/(mA)$ grows with the momentum J; as $J \to 0$ the barrier disappears (annihilation).

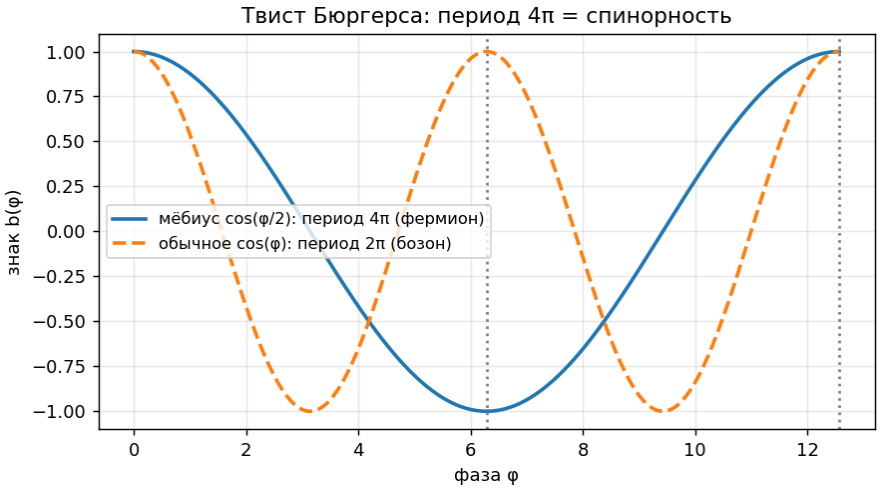


Figure 3: Burgers twist: $\cos(\varphi/2)$ is antiperiodic under 2π and returns only after 4π (spinor, fermion), unlike the ordinary $\cos\varphi$ with period 2π (boson).